

LO4: Gaps and Omissions

Although testing carried out was extensive, covering most functional, safety and performance requirements, it did not allow for full certainty that all requirements held in all cases. Test adequacy criteria can only act as proxies for correctness and do not guarantee no faults, they are mainly useful for identifying inadequacies.

LO4.1: One limitation of the testing process became apparent through a defect that was initially missed. Although the path generation logic correctly followed movement constraints and satisfied performance requirements, an unintended extra hover was implemented at each stop. This behaviour was difficult to detect through informal testing as the path generated was still technically valid, but it is a mistake that affected the total cost over time, making it a significant fault in respect to the cost requirement.

This defect was not originally discovered in testing, revealing a gap in the testing of cost-related behaviour. In particular, although movement correctness and safety were tested thoroughly, the relationship between movement steps, hover behaviour and cost calculation was not explicitly verified. The fault was later found in marking, showing the current tests were not enough to guard against this type of error.

After discovering this issue, an additional test should be added to explicitly check that the number of hovers matched the expected amount. This shows how evaluation of test outcomes can expose omissions in test coverage and motivate targeted improvements.

LO4.2: For functional requirements such as movement constraints and restricted-area avoidance, the target was to achieve coverage of all identified equivalence classes and boundary conditions derived from the specification. For safety critical behaviour, the implicit target was that no test execution should result in a violation of no-fly zones.

For measurable attributes, the primary target was compliance with the explicit performance requirement that route computation must complete within 30 seconds. Cost calculation accuracy was also an implicit target, although this was treated as secondary to safety and performance where it shouldn't have been.

These targets are realistic within the scope of the project but remain weaker than full correctness guarantees, particular for cumulative properties such as cost.

LO4.3: Testing largely met the stated targets for functional correctness, safety and performance. All safety related tests passed, and performance testing demonstrated compliance with the 30 second execution time limit under controlled workloads. However, the missed hover related cost defect shows that the achieved test coverage was insufficient to fully meet the intended cost related targets.

This demonstrates a key limitation: satisfying commonly used criteria such as equivalence class coverage and branch coverage does not guarantee that all important behaviours have been tested, especially where requirements involve cumulative or derived values.

LO4.3: To better achieve the target levels, the testing process would need to be extended in several ways. Additional cost-focused tests should be introduced that explicitly relate movement sequences to

total cost, rather than treating cost as a secondary outcome. Structural coverage could be measured using automated tools to identify untested paths related to cost accumulation.

More generally, fault-based thinking could be applied by hypothesising common classes of errors, such as mistakes that are off by one or unintended repeated actions, and designing tests specifically to reveal them. While resource constraints limit how far this can be taken, incorporating such tests would improve confidence that the system satisfies both its functional and measurable requirements.